

Man-In-The-Middle Attacks

Definition

- Man-in-the-Middle (MitM) attacks happen when traffic between two parties is observed or manipulated by an unknown third party.
- A MitM attack is a cybercrime method used to steal personal information or login credentials. Cyber criminals also use MitM attacks as a means to spy on, corrupt information, or disrupt communications between two parties.

Since the 1980s, MitM attacks have been used to infiltrate traffic between innocent parties.

Methods

Man-in-the-Middle attacks can happen in a number of ways:

- Types of Spoofing (IP, DNS, HTTPS)
- Hijacking (Secure Socket Layer, Email)
- Wi-Fi Eavesdropping
- Theft of Browsing Cookies

Man-in-the-Browser:

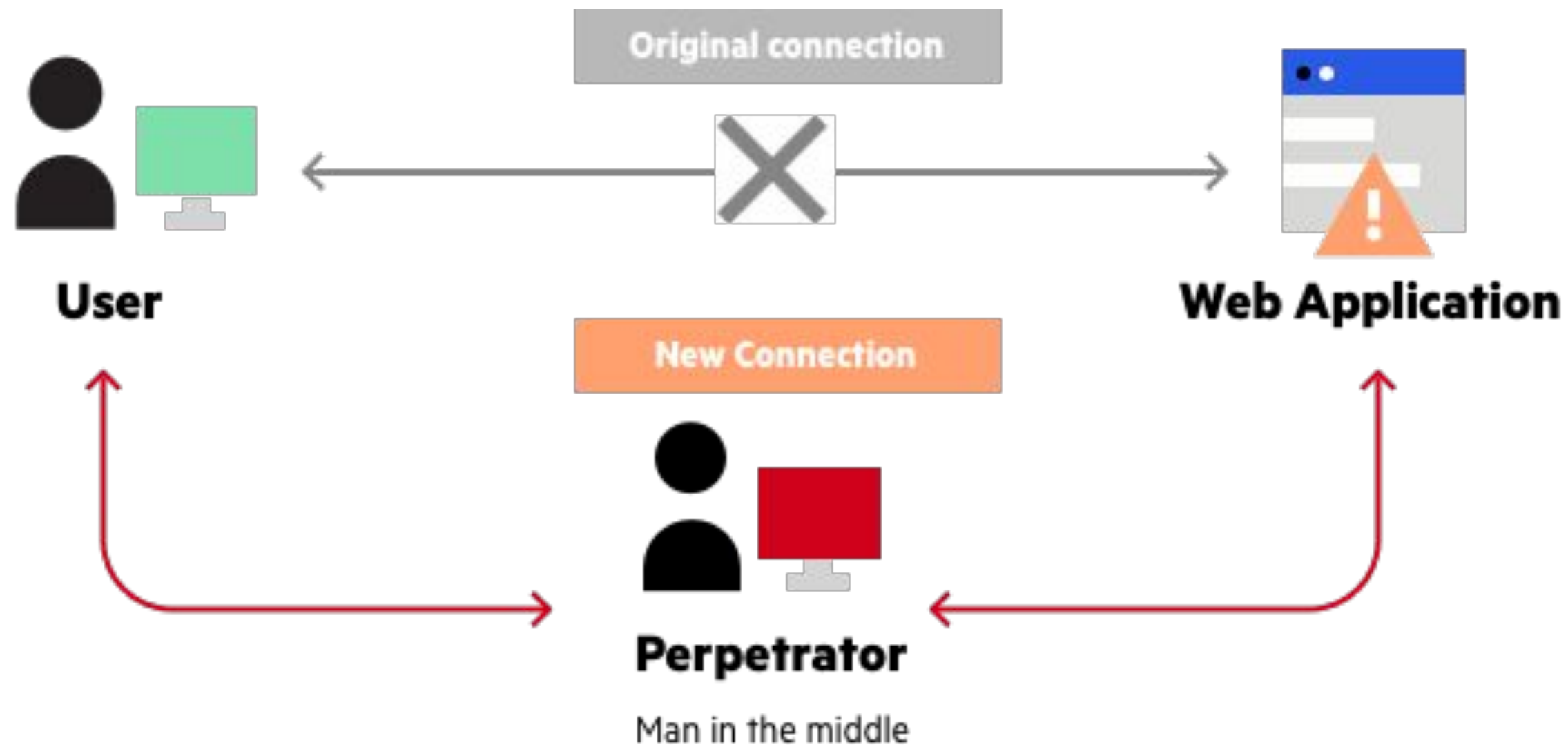
- When a Man-in-the-Browser attack takes place, the attacker uses a Phishing method in order to administer malware to a device.
- Malware is software meant to damage a network, server or personal computer.

Phishing is a method of sending fraudulent emails or text messages to trick a user into revealing personal information.

Man-in-the-Browser Continued

- A Man-in-the-Browser attack happens when malware installs itself on a victim's browser in order to record information sent between targeted websites and the user.
- Online banking institutions are prone to this form of cybercrime.

Example of Man-in-the-Browser

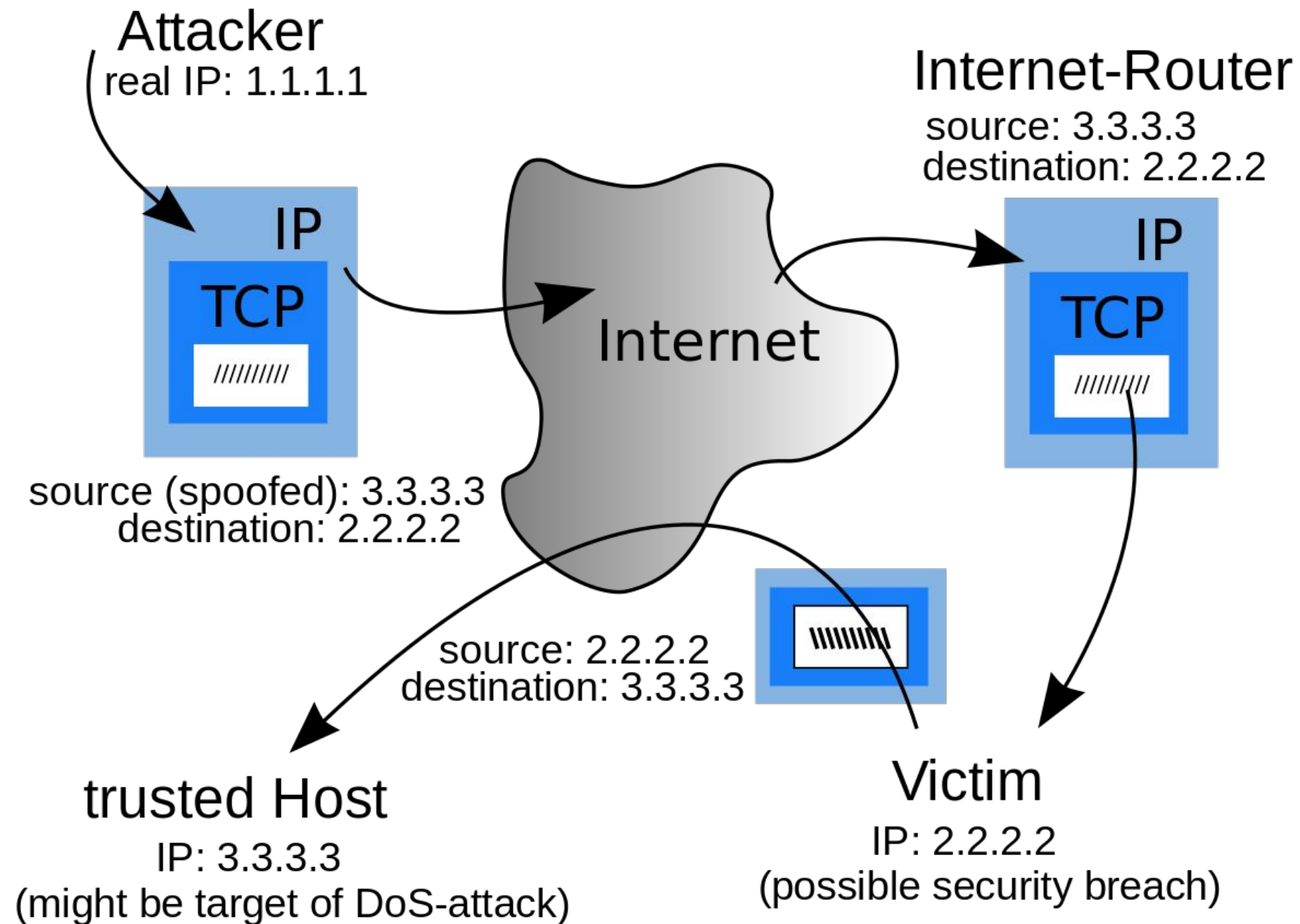


IP Spoofing

An IP Address is like your home address.

- All devices that connect to the internet have an IP Address.
- Spoofing happens when someone or something impersonates a trusted source.
- Attackers use IP Spoofing in order to deceive users into revealing sensitive information by “spoofing” their IP and posing as a website or someone familiar.

Example of IP Spoofing

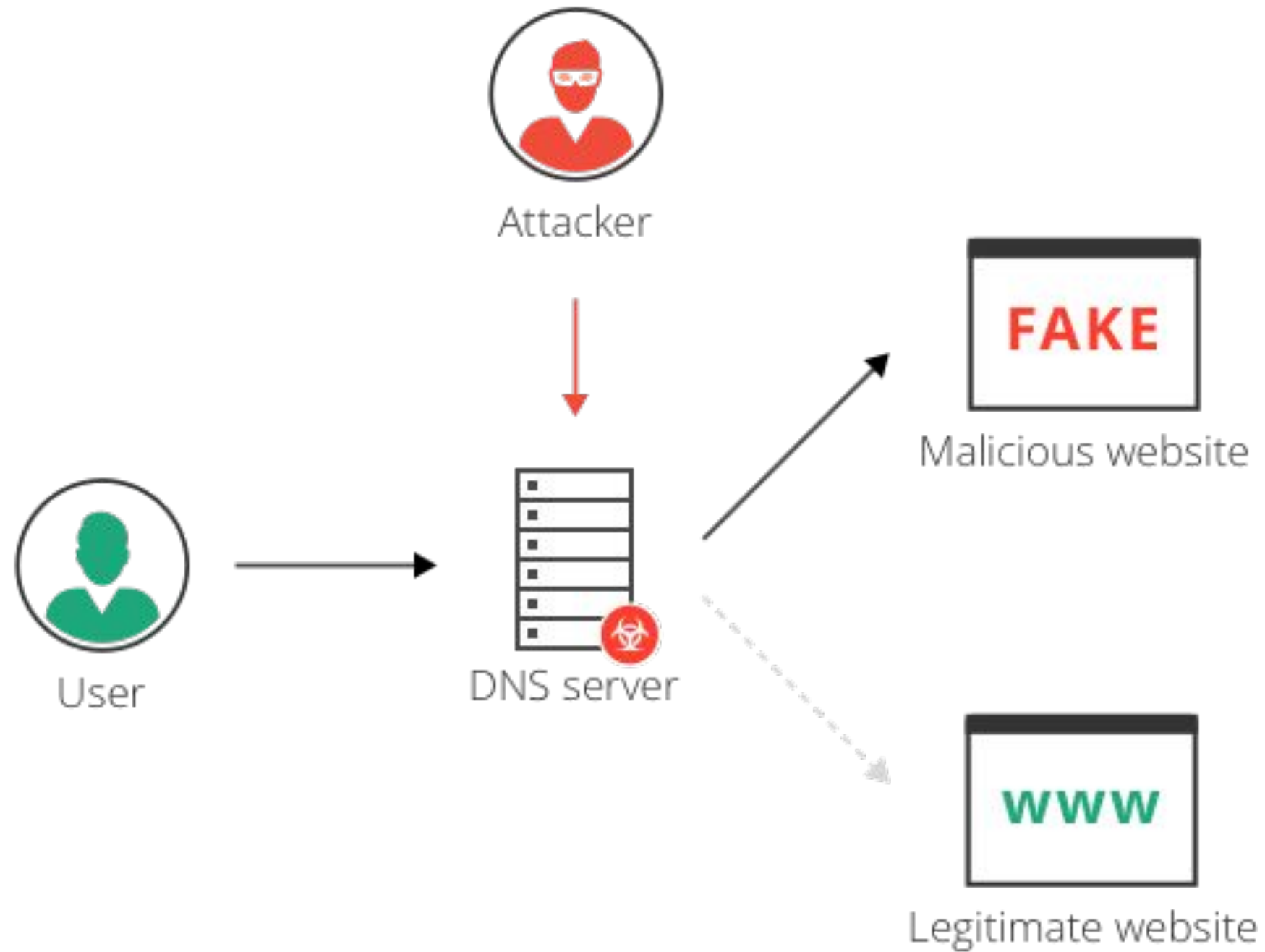


DNS Spoofing

- DNS refers to “Domain Name Server/System”. The DNS system converts names to IP Addresses.
- When Spoofing a DNS, a user is forced to an imitation website, similar to the one intended to be viewed.
- The goal of the attacker is to divert traffic or retrieve login credentials.

Example: The DNS will return the IP address of a visited website when it is typed into a browser.

Example of DNS Spoofing



HTTPS Spoofing

- HTTPS stands for Hypertext Transfer Protocol Secure. This protocol is used by the Web to format and transmit messages.
- Ensure that when visiting a website, the URL indicates that it uses “https” rather than “http”.

HTTPS Spoofing

- In HTTPS Spoofing, the attacker manipulates the browser into believing it is visiting a trusted website.
- After redirecting your browser, the attacker uses the vulnerable website to monitor communication and steal shared information.

SSL Hijacking

- SSL stands for Secure Sockets Layer, which was a protocol developed in order to communicate over the internet securely.
- Sometimes when a device visits an unsecure website (http), it is automatically redirected to the secure version (https).
- An attacker utilizes a computer and secure server to reroute information of a user right before connection to a legitimate server, this is SSL Hijacking.

Email Hijacking

- Email Hijacking occurs when attackers target financial organizations for email information.
- After obtaining access to email accounts, attackers can monitor all financial transactions.
- Attackers then follow up by “spoofing” the financial institution’s email and possibly providing users with instructions that would result in the attacker receiving funds.

Wi-fi Eavesdropping

- Wi-Fi connections can be configured and appear to have a valid name, such as the Wi-fi of a favourite coffee shop.
- If a user connects to the fraudulent Wi-Fi connection, the user's online activities can be observed and personal information like banking cards can be attained.

Precautions should be taken when connecting to public Wi-Fi connections.

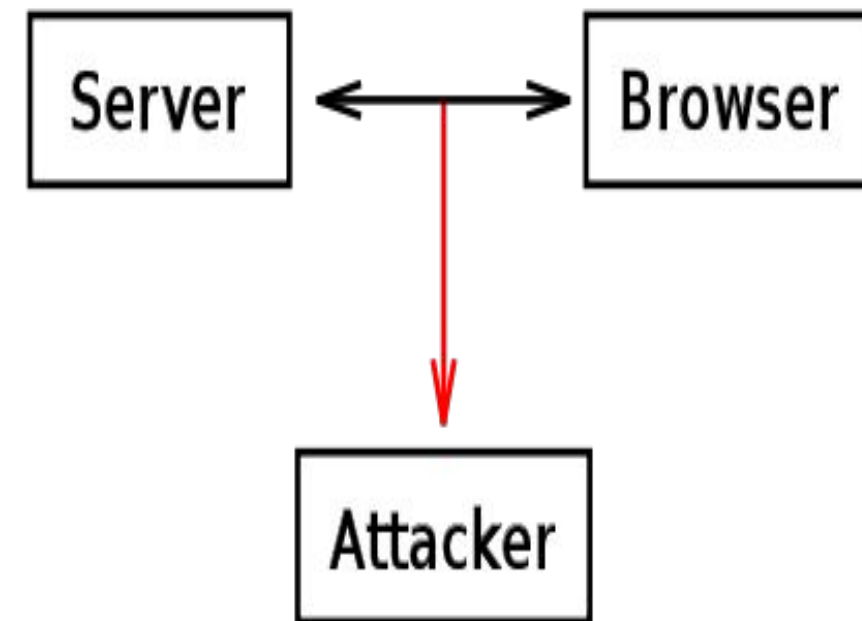
Example of Wi-fi Eavesdropping



Precautions should be taken when connecting to public Wi-Fi connections.

Browser Cookies

- A browser cookie is a small piece of data stored by the user's web browser. This data is used to track browsing sessions.
- If browsing data is stored in a cookie and the browser cookie is hijacked, cybercriminals may be able to gain passwords, addresses and other sensitive information.



Protection

- Ensure that the browser is using “https” when browsing the web.
- Be on alert of Phishing emails that request credentials to be updated.

Protection Continued

- Refrain from connecting to public Wi-Fi connections without a VPN.
- Make use of internet security applications to thwart MitB attacks.